

KMOU Division of Marine System Engineering

Introduction to Internal Combustion Engine



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Lesson content for each week

Ch0_Introduction to the Class

Ch6_Piston Fittings and Crankshaft

Ch1_Overview of the Internal Combustion Engine

Ch7_Lube Oil System

Ch2_Thermodynamics and Compressed Air System

Ch8_Cooling Water System

Ch3_Marine Fuels and Internal Fuel Oil System

Ch9_Introduction to the Main Engine

Ch4_Intake and Exhaust System

Ch10_Alternative Fuels and Alternative Fuel Systems

Ch5_Turbocharger

Ch11_Environmental Regulations and Exhaust Gas Aftertreatment Systems

Midterm

Ch12_Tools, Apprentice Engineer Duties, and Career Direction

Final Exam

What is Turbocharger (T/C)?

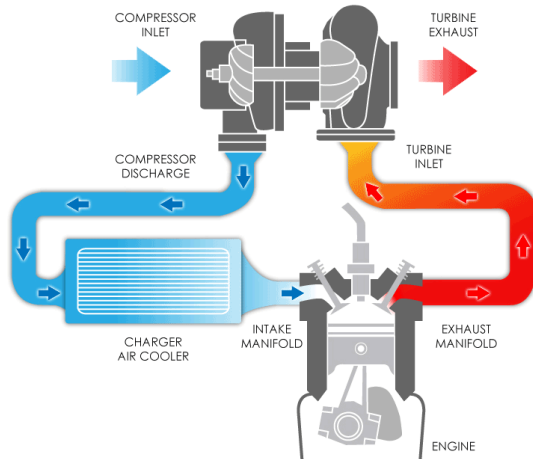
Most of the G/E turbochargers used on ships are made by ABB.



Therefore, this chapter explains ABB's T/C.

T/Cs are intended for turbocharging internal combustion engines.

The T/C is used to generate the volume of air and the charging pressure required to operate the engine.



Turbocharger (T/C)

The T/C is a turbomachine consisting of the following main components:

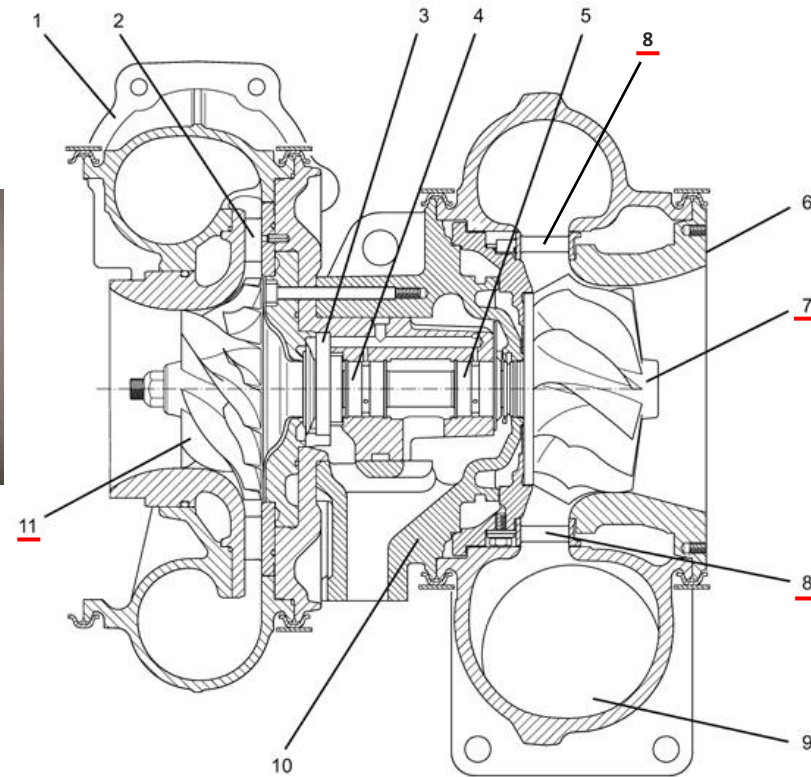
- turbine
- compressor

These are mounted on a common shaft.

The exhaust gas from the diesel or gas engine flows through the gas inlet casing (9) and nozzle ring (8) to the turbine (7).

The turbine (7) uses the energy contained in the exhaust gas to drive the compressor wheel (11). The compressor draws in fresh air, compresses it and then forces it into the engine's cylinders.

Nozzle ring



- 1 Compressor casing
- 2 Diffuser
- 3 Axial thrust bearing
- 4 Radial bearing
- 5 Radial bearing
- 6 Gas outlet flange

- 7 Turbine
- 8 Nozzle ring
- 9 Gas inlet casing
- 10 Bearing casing
- 11 Compressor wheel

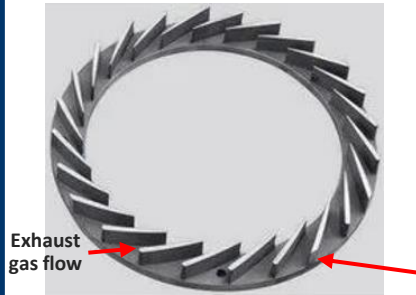
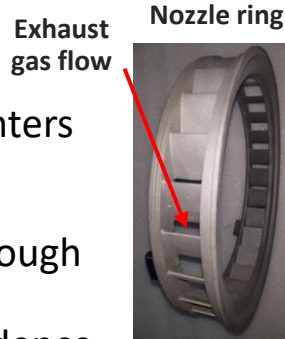
Turbocharger (T/C)

Nozzle ring

It is located just before the exhaust gas enters the turbine wheel.

The flow of exhaust gas is accelerated through the nozzle vane (blade) and guided to the turbine blade at the optimal angle of incidence.

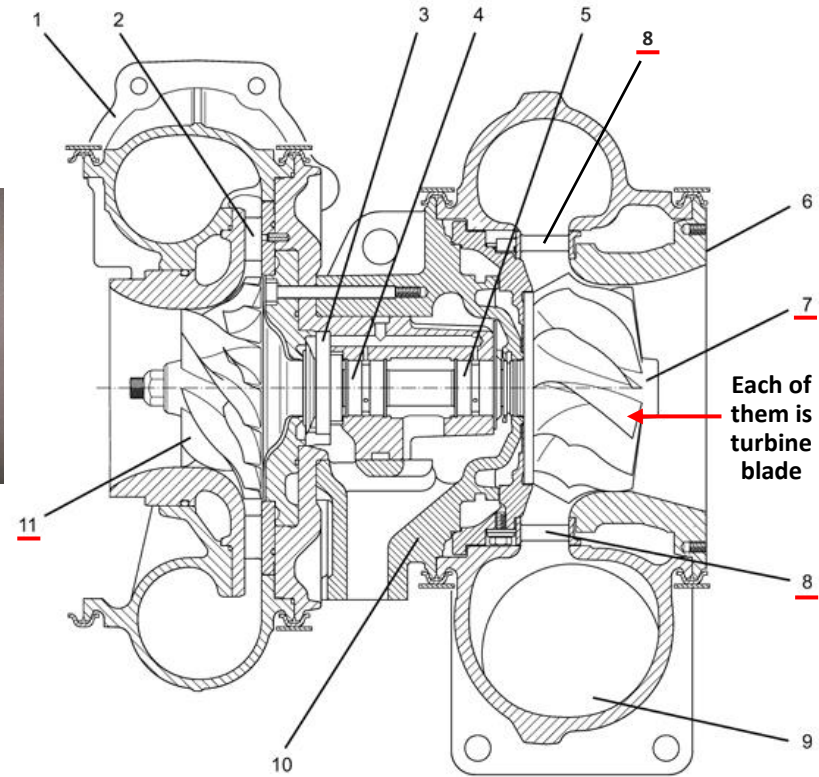
It converts the pressure energy of the exhaust gas into kinetic energy to increase turbine efficiency.



The nozzle increases the velocity of the fluid by making the outlet cross-sectional area smaller (contracting) than the inlet.

It is called a nozzle ring because it has multiple nozzles in a ring shape.

Each of them is nozzle vane (blade)



- 1 Compressor casing
- 2 Diffuser
- 3 Axial thrust bearing
- 4 Radial bearing
- 5 Radial bearing
- 6 Gas outlet flange

- 7 Turbine
- 8 Nozzle ring
- 9 Gas inlet casing
- 10 Bearing casing
- 11 Compressor wheel

Turbocharger (T/C)

V: Volume

v: Velocity

Bernoulli's equation

Energy per unit volume before = Energy per unit volume after

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2$$

Pressure Energy Kinetic Energy per unit volume Potential Energy per unit volume

Flow velocity v_1 Flow velocity v_2

The often cited example of the Bernoulli Equation or "Bernoulli Effect" is the reduction in pressure which occurs when the fluid speed increases.

$A_2 < A_1$
 $v_2 > v_1$
 $P_2 < P_1$!

Increased fluid speed, decreased internal pressure.

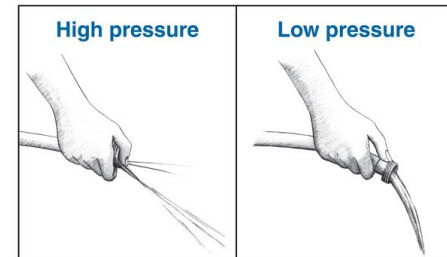
Continuity equation

Section 1 Section 2

A_1 v_1 L_1 A_2 v_2 L_2

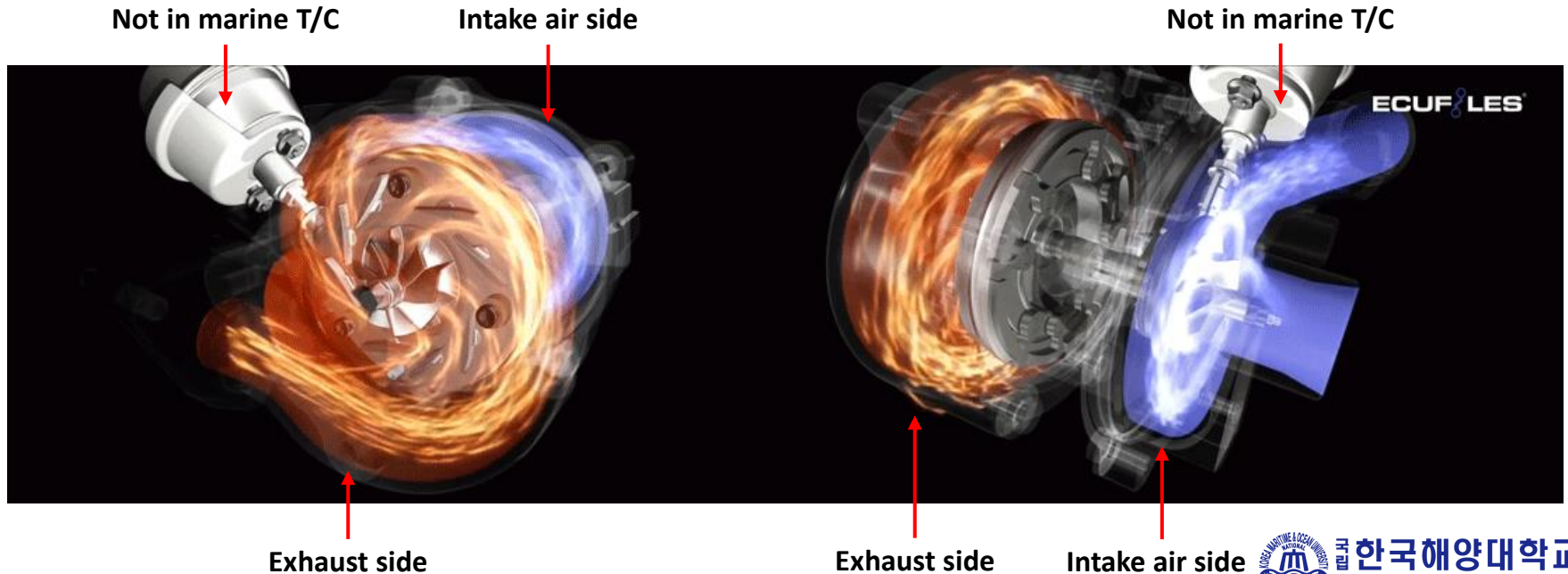
$$Q_1 = V_1/t$$
$$Q_1 = (A_1 L_1) / t$$
$$Q_1 = A_1 v_1$$
$$Q_2 = A_2 v_2$$
$$Q_1 = Q_2$$
$$A_1 v_1 = A_2 v_2$$

Continuity Equation

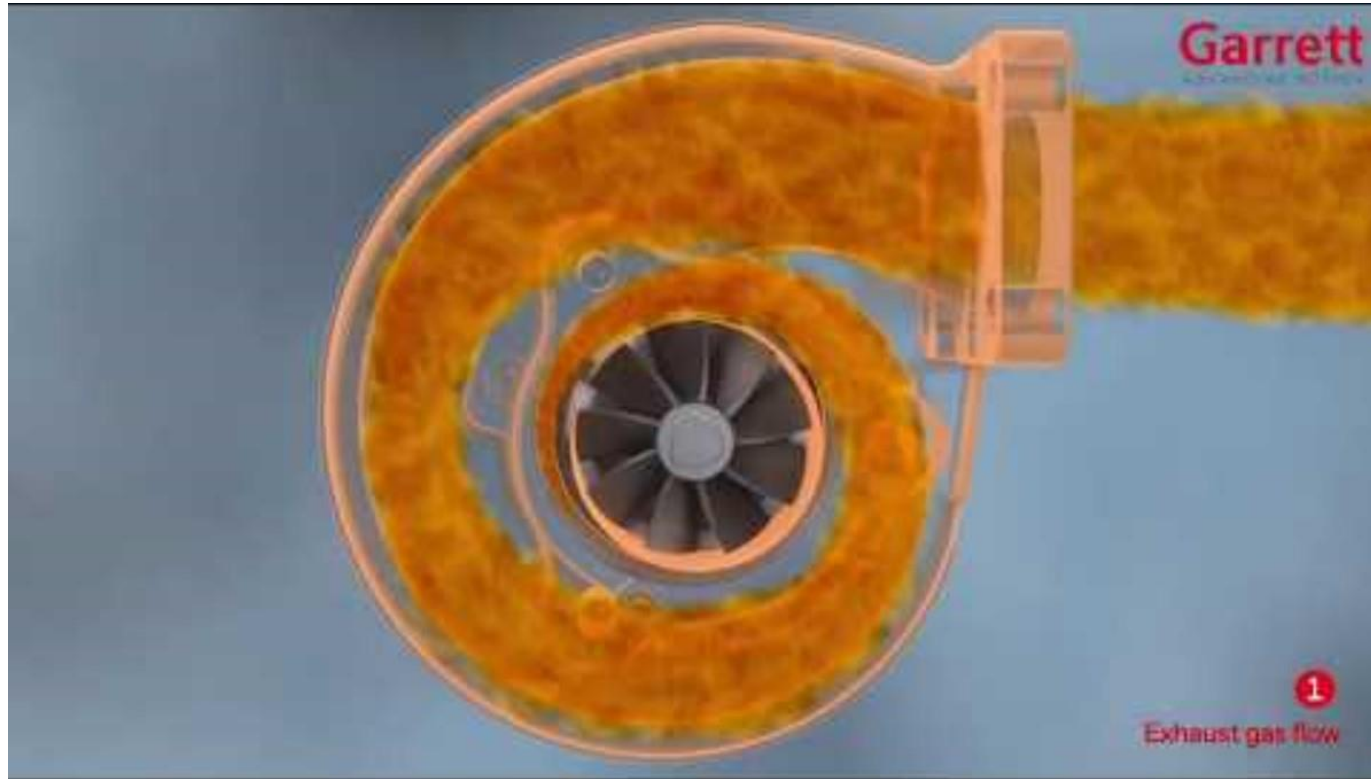


Turbocharger (T/C)

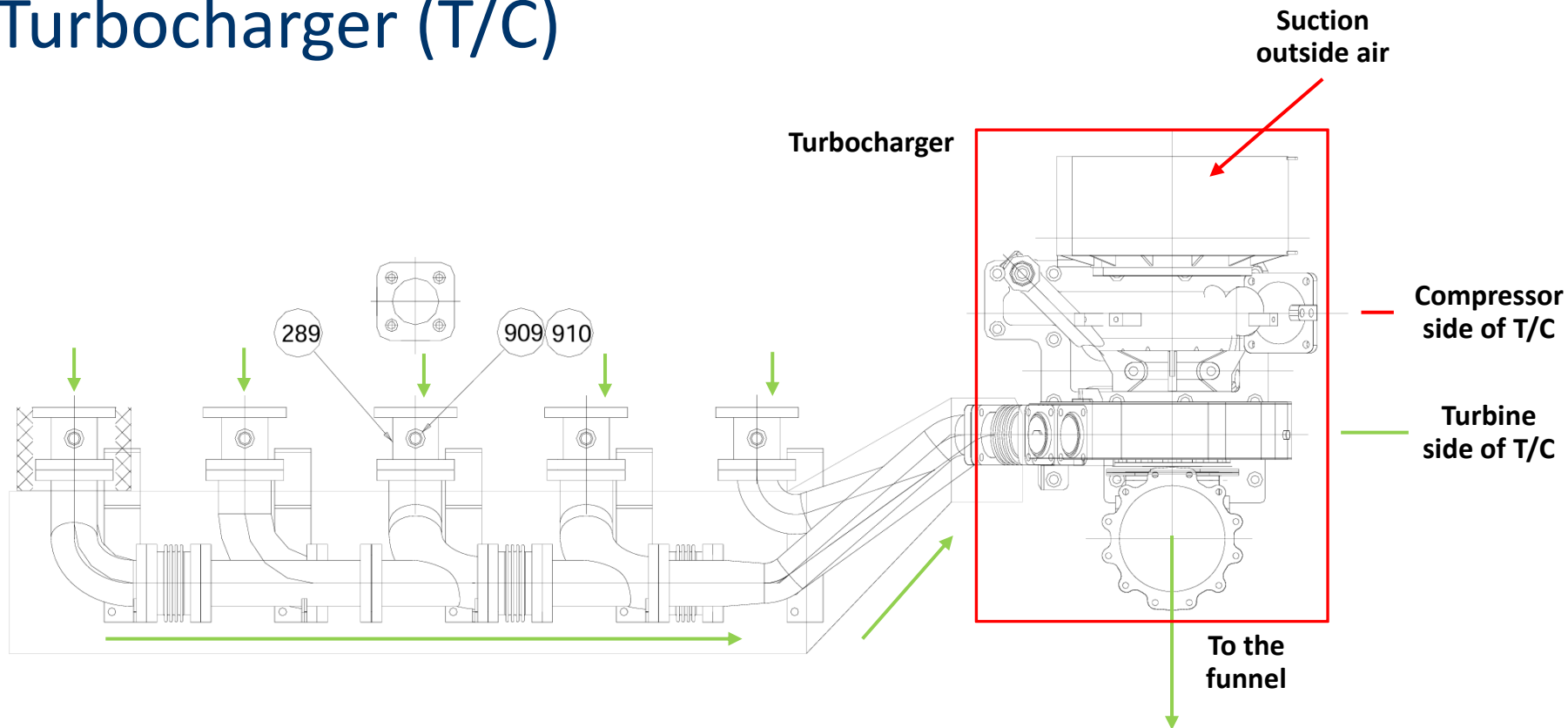
This is just an animation to illustrate the exhaust gas passing through the nozzle ring to the turbine wheel (left) and the fresh air being compressed by the compressor wheel (right).



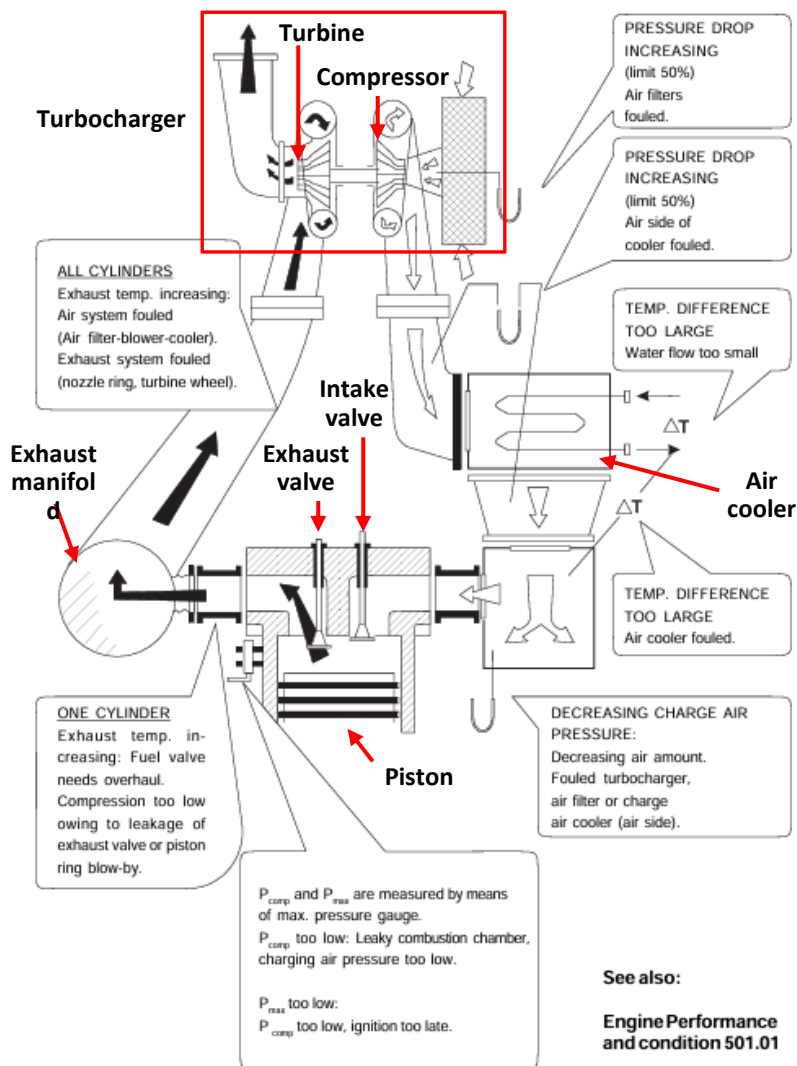
Turbocharger (T/C)



Turbocharger (T/C)



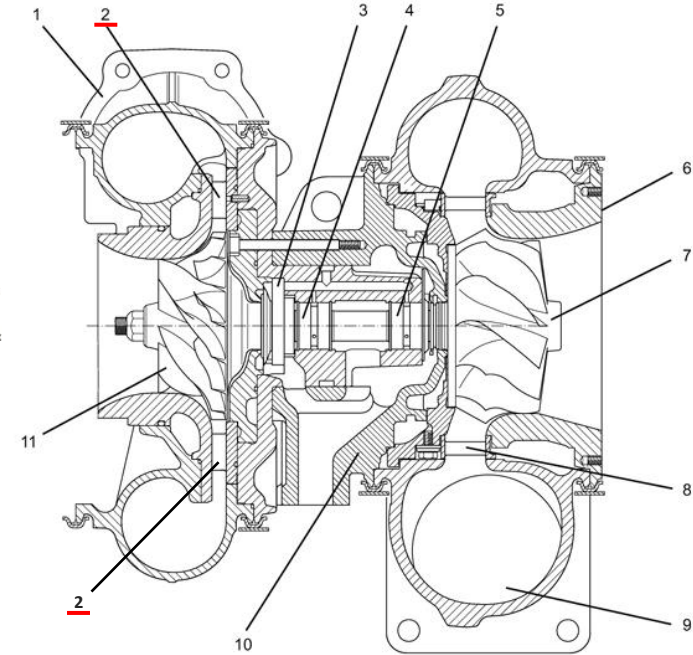
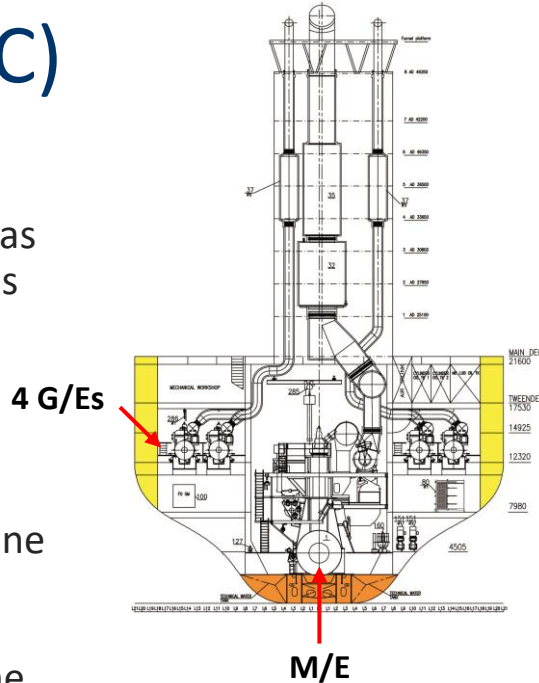
Turbocharger (T/C)



Turbocharger (T/C)

The exhaust gas escapes into the ambient air through the exhaust gas pipe, which is connected to the gas outlet flange (6).

The air, which is necessary for operation of the diesel or gas engine and which is compressed in the turbocharger, is drawn into the compressor wheel (11) through the suction branch or filter silencer or through an axial suction pipe. It then passes through the diffuser (2) and leaves the turbocharger through the outlet on the compressor casing (1).



- 1 Compressor casing
- 2 Diffuser
- 3 Axial thrust bearing
- 4 Radial bearing
- 5 Radial bearing
- 6 Gas outlet flange

- 7 Turbine
- 8 Nozzle ring
- 9 Gas inlet casing
- 10 Bearing casing
- 11 Compressor wheel



Turbocharger (T/C)

Diffuser

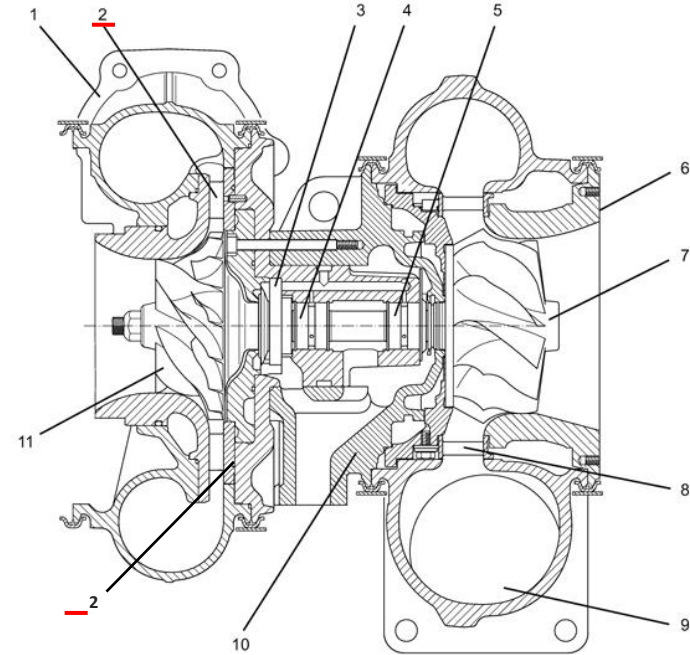
It is located between the compressor wheel outlet of the compressor and the compressor housing.

It converts the kinetic energy of the high-speed air exiting the compressor wheel into static pressure by slowing it down. In other words, the main pressure-increasing section of the compressor is the diffuser.

It increases the compressor discharge pressure to supply more air to the engine.

The diffuser has the same arrangement of vanes as the nozzle ring, with the outside being wider and the inside being narrower.

However, in the case of the nozzle ring, the exhaust gas enters from the outside, so the velocity increases, and in the case of the diffuser, the intake air exits from the inside to the outside, so the pressure increases.



- 1 Compressor casing
- 2 Diffuser
- 3 Axial thrust bearing
- 4 Radial bearing
- 5 Radial bearing
- 6 Gas outlet flange

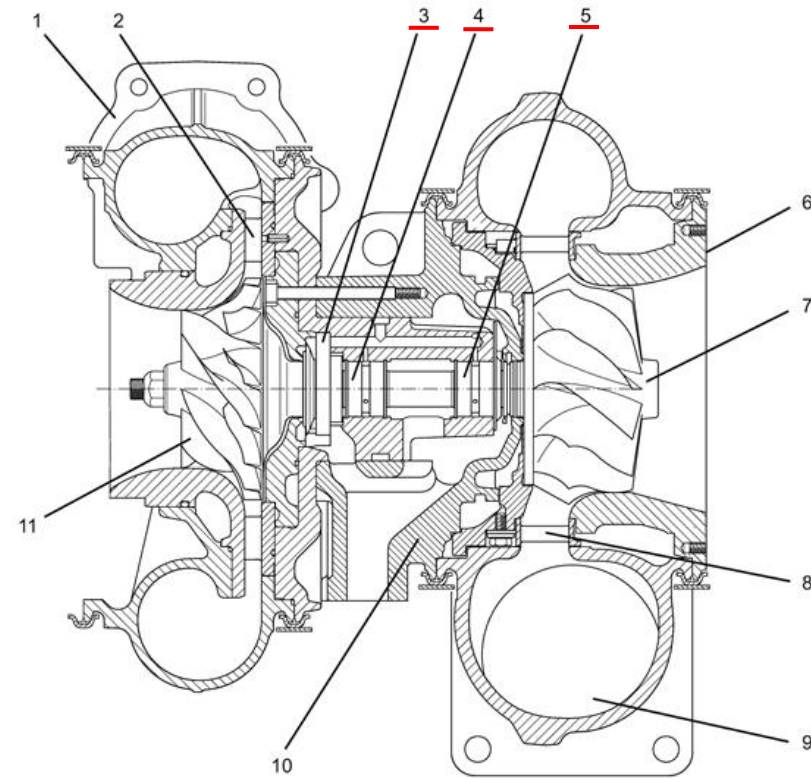
- 7 Turbine
- 8 Nozzle ring
- 9 Gas inlet casing
- 10 Bearing casing
- 11 Compressor wheel



Turbocharger (T/C)

The rotor runs in two radial bearings (4/5) which are located in the bearing flange between the compressor and the turbine. The axial thrust bearing (3) is located at the compressor end.

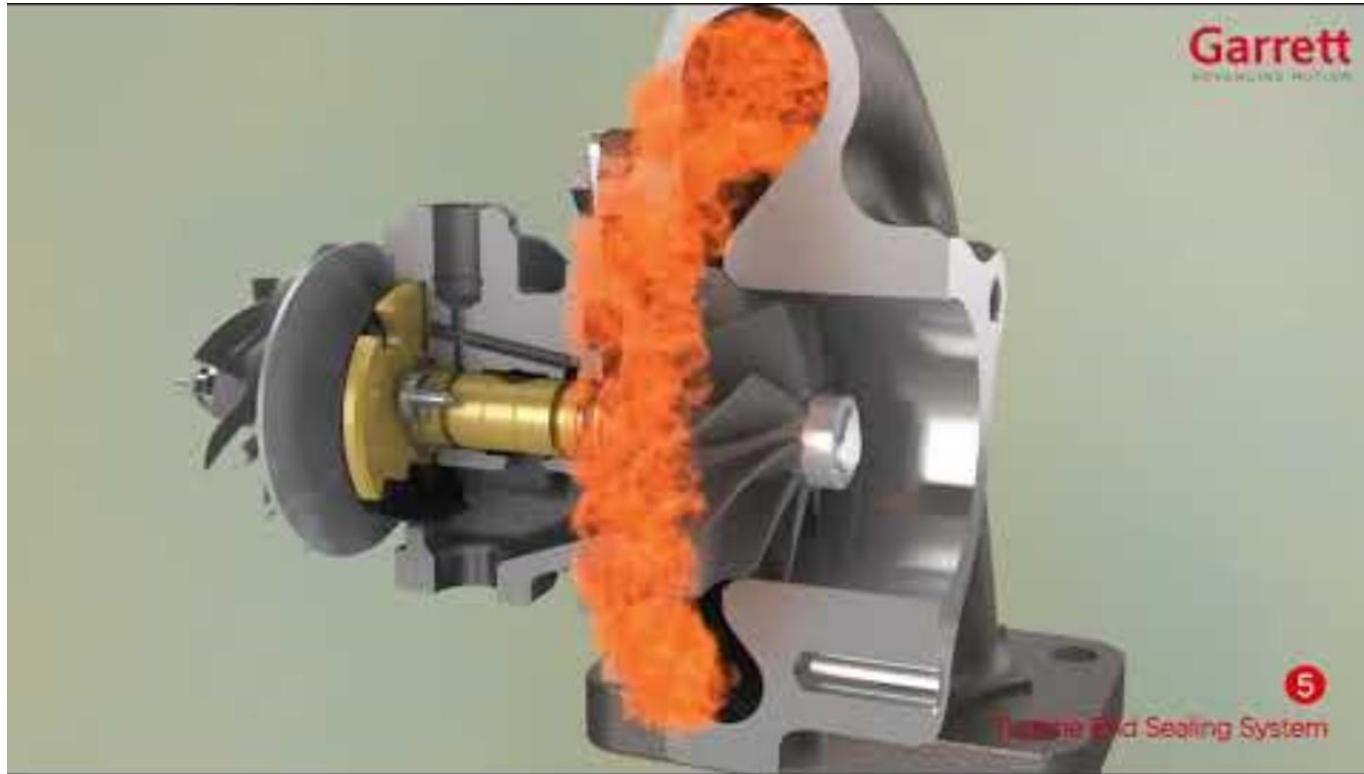
The radial bearings are connected to a central lubricating oil duct which is fed with oil from the engine's lubricating oil circuit. The oil outlet is always at the lowest point of the bearing casing (10).



- 1 Compressor casing
- 2 Diffuser
- 3 Axial thrust bearing
- 4 Radial bearing
- 5 Radial bearing
- 6 Gas outlet flange

- 7 Turbine
- 8 Nozzle ring
- 9 Gas inlet casing
- 10 Bearing casing
- 11 Compressor wheel

Lubricant flow in the Turbocharger (T/C)



Turbocharger (T/C)

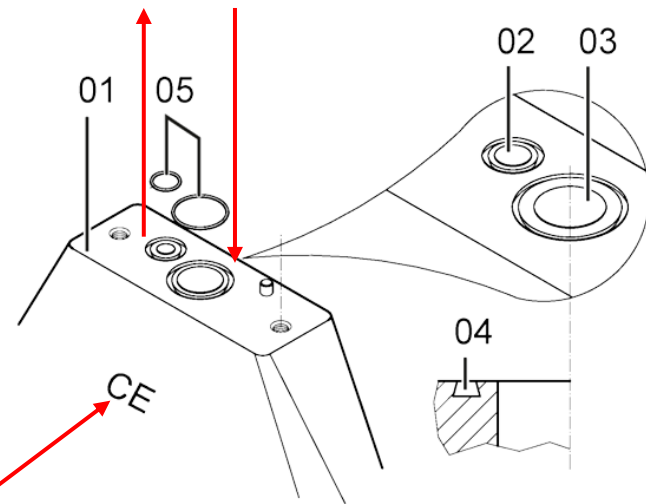
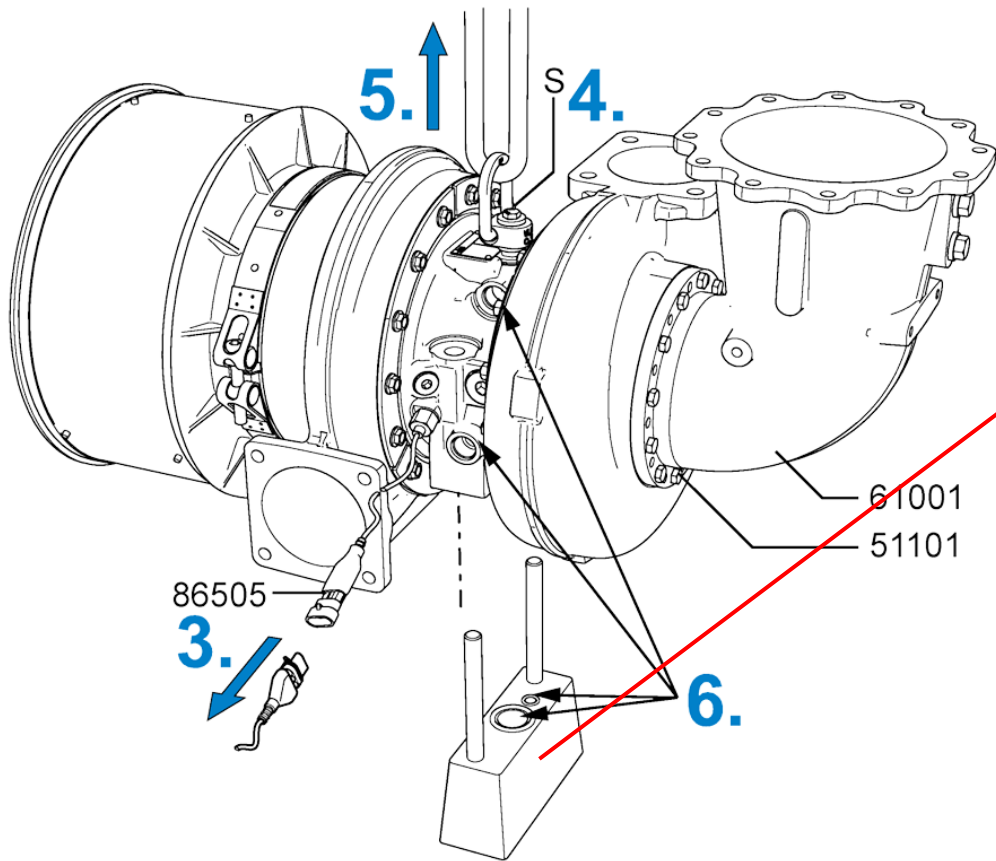


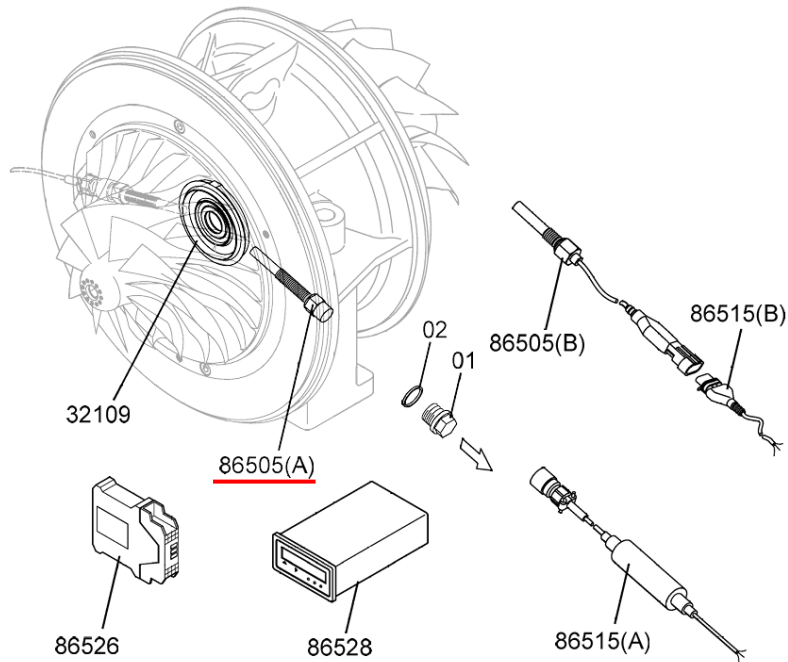
Fig. 6: Inserting O-rings into bracket

- | | |
|-----------|------------------|
| 01 | Bracket |
| 02 | Oil supply |
| <u>03</u> | <u>Oil drain</u> |
| 04 | Slot for O-ring |
| 05 | O-rings |
| CE | Compressor end |

Turbocharger (T/C)

Speed measurement

A speed measuring system permits continuous monitoring of the turbocharger speed.



86505(A) Speed sensor (cable not integrated)

86505(B) Speed sensor (cable integrated)

86515(A) Cable connector to 86505(A)

86515(B) Cable connector to 86505(B)

86526 F/I converter

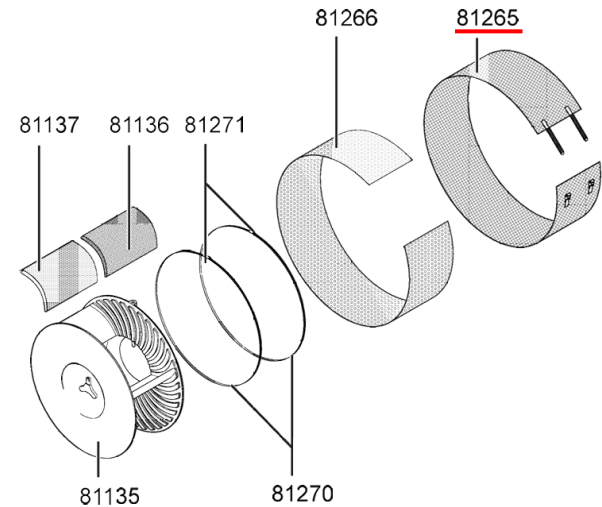
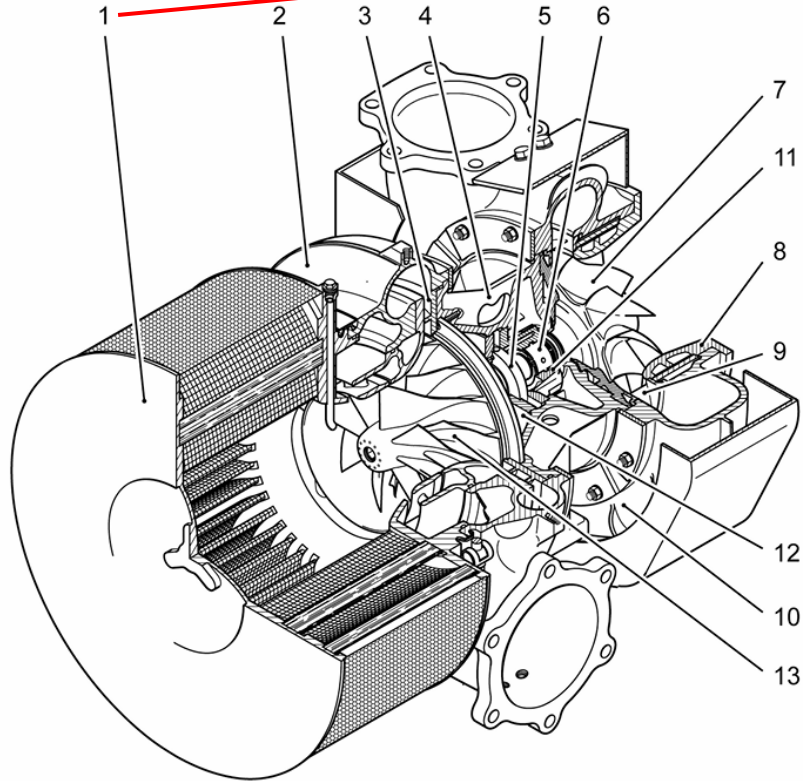
86528 Tachometer

01 Screw plug

02 Gasket

Components of T/C

Layout and function



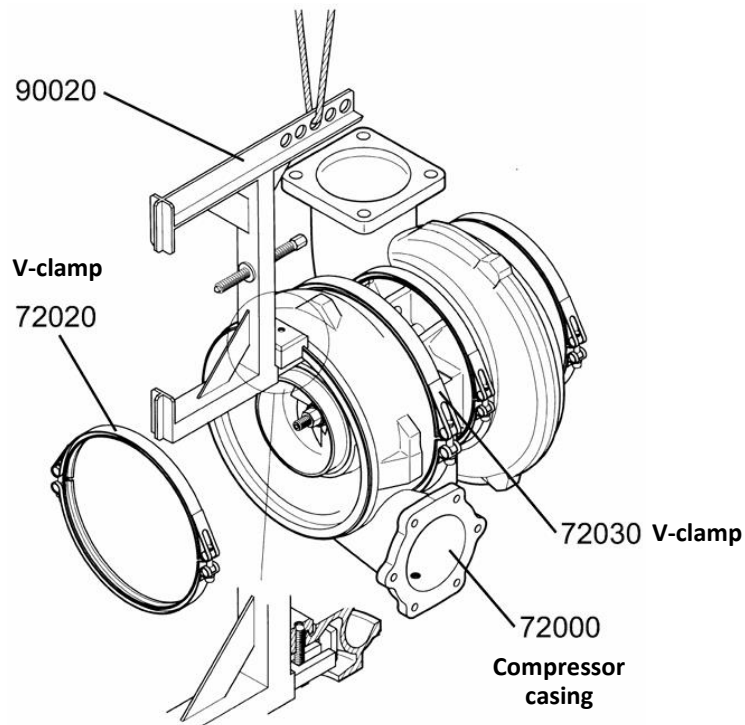
81135 Filter silencer body
81136 Absorption segment
81137 Sheet-metal covering
81265 Filter ring

81266 Cover grid
81270 Tension band
81271 Lock

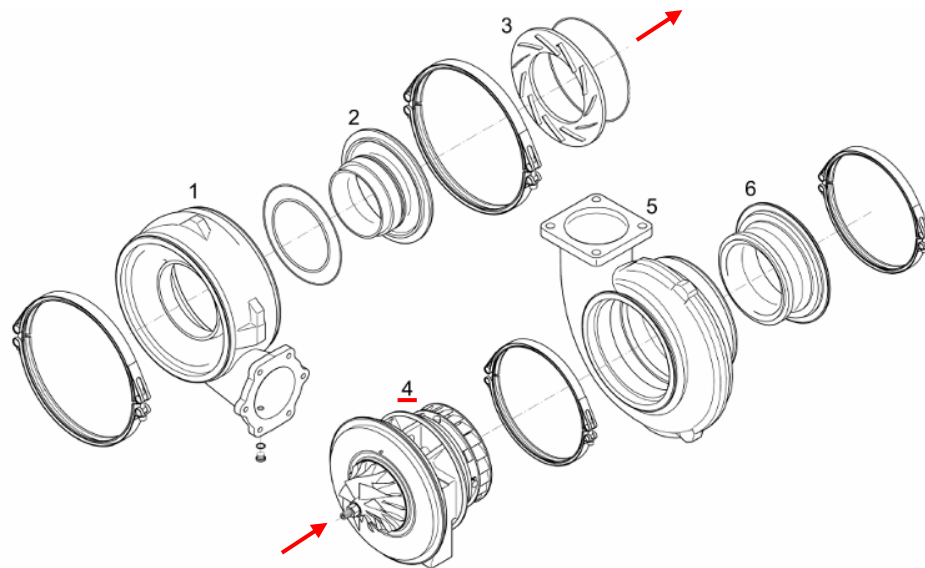
1 Suction branch / filter silencer
2 Compressor casing
3 Diffuser
4 Bearing casing
5 Axial thrust bearing
6 Radial plain bearing (visible only
at turbine end)
7 Turbine

8 Gas outlet flange
9 Nozzle ring
10 Turbine casing
11 Turbine-end bearing flange
12 Compressor-end bearing flange
13 Compressor wheel

Components of T/C



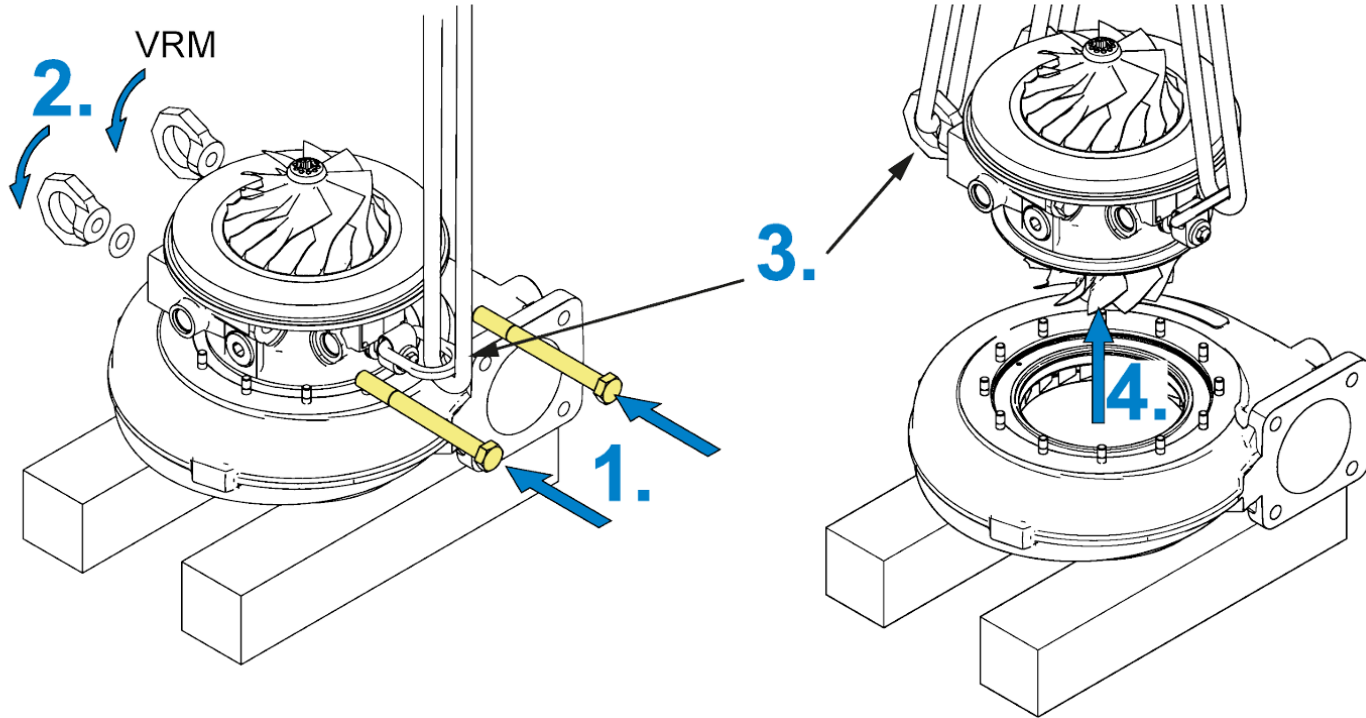
Assemblies



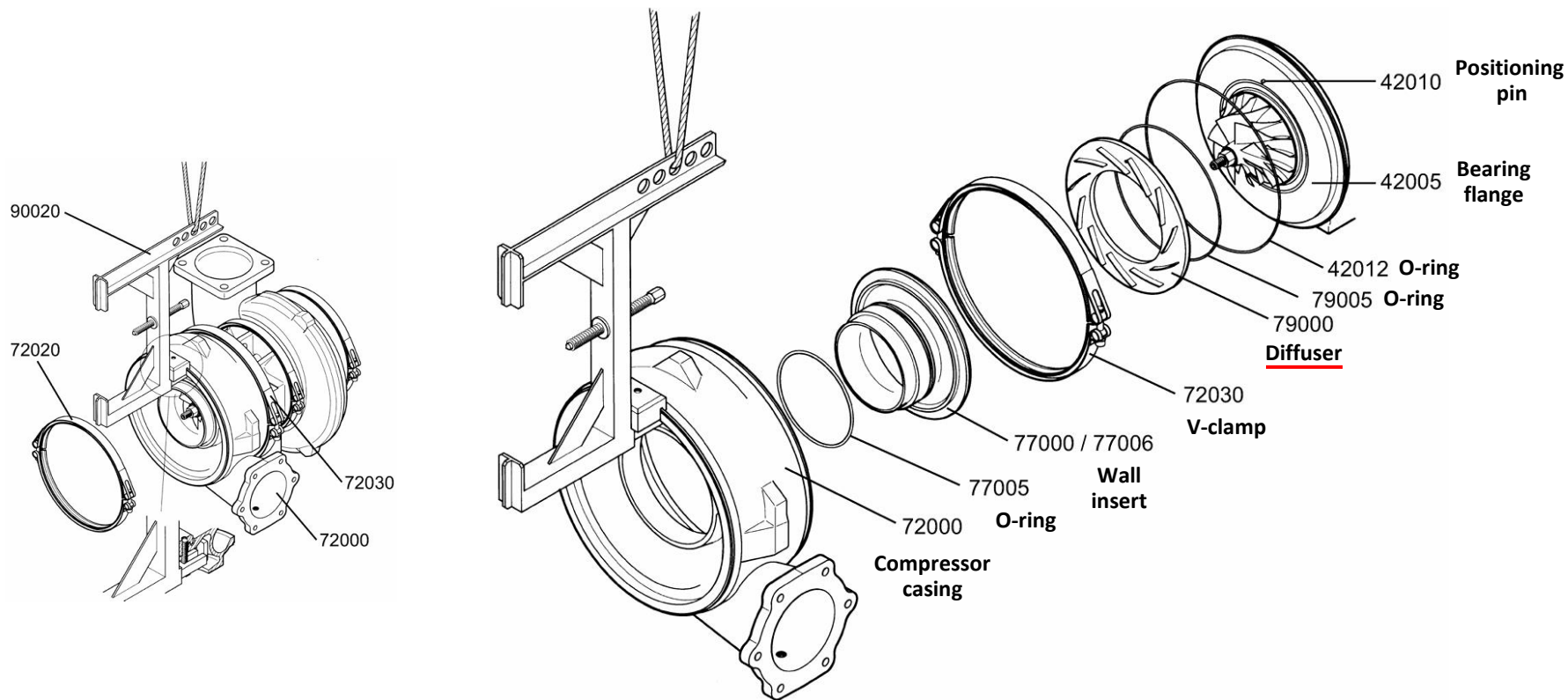
	Description	TPS 50
1	Compressor casing	55
2	Wall insert	13
3	Diffuser	5
4	<u>Cartridge group</u>	85
5	Turbine casing	65
6	Gas outlet flange	23

Components of T/C

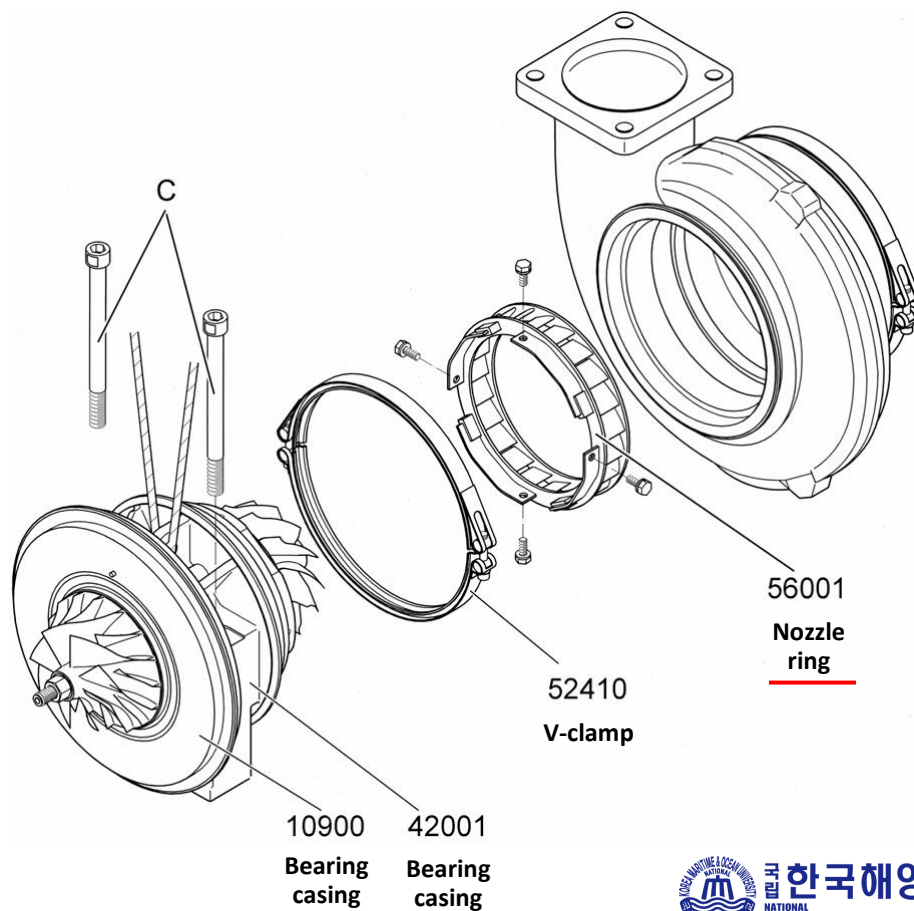
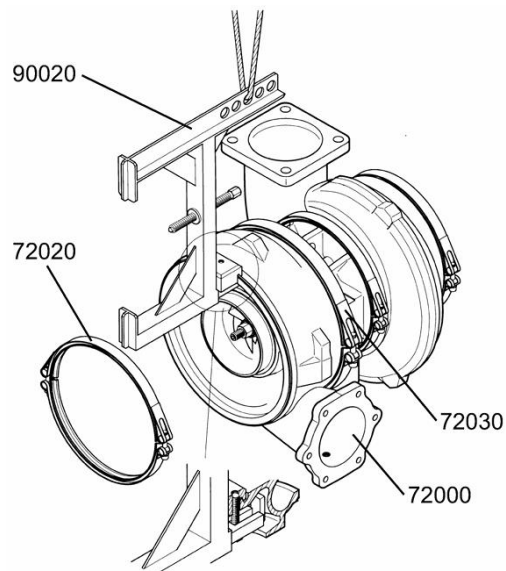
Removing the cartridge group



Components of T/C



Components of T/C

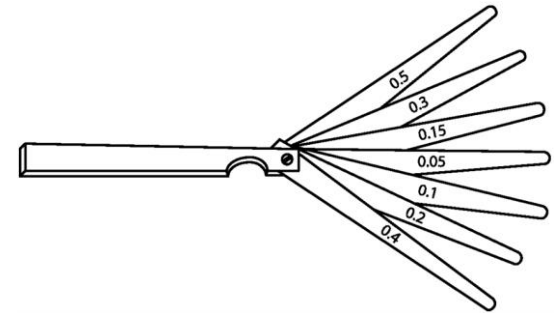
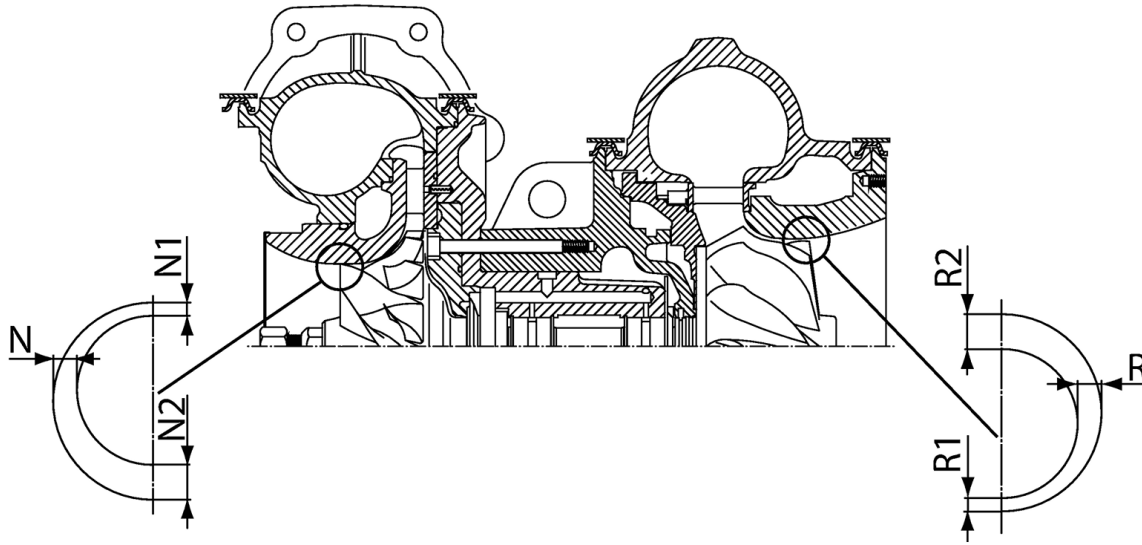


Radial clearances N and R for T/C

These are the theoretical values of the radial clearance between the compressor wheel and the wall insert and / or turbine wheel and the gas outlet flange.

The minimum and maximum values result from the manufacturing tolerances of the compressor wheel and turbine wheel or the wall insert and gas outlet flange.

N1 and N2 as well as R1 and R2 are each measured at the same time using two feeler gauges.

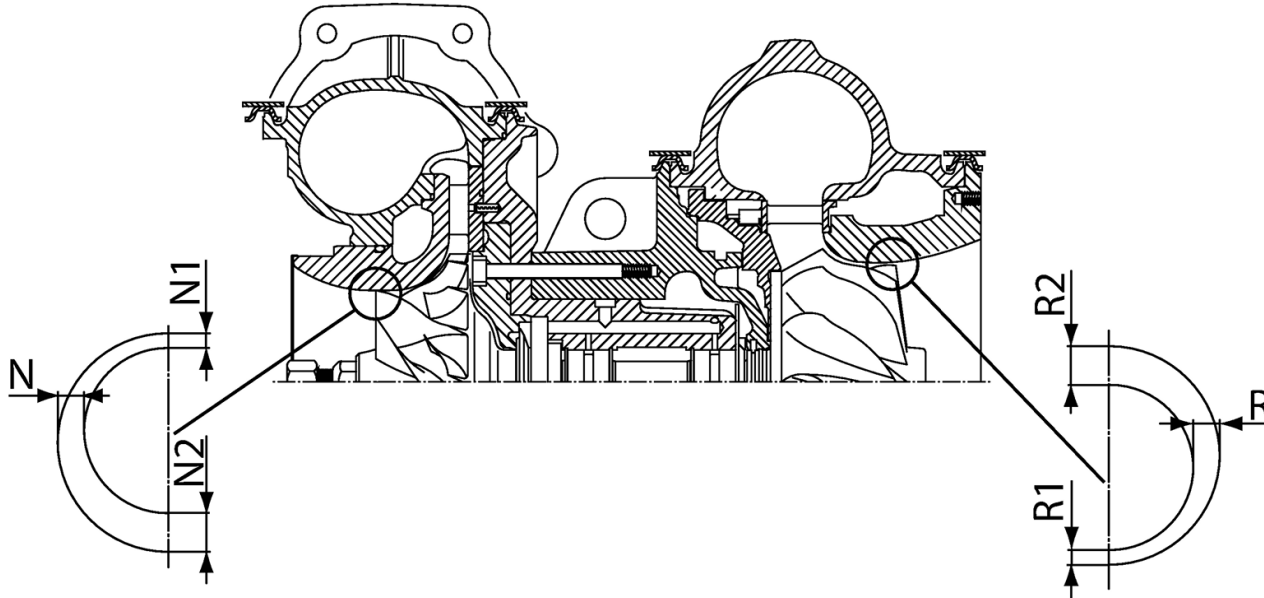


Feeler gauge

Radial clearances N and R for T/C

To measure N1 and N2, the feeler gauges must be pushed between the wall insert and the compressor wheel free of play, in each case above and below.

To measure R1 and R2, the feeler gauges must be pushed between the gas outlet flange and the turbine free of play, in each case above and below.



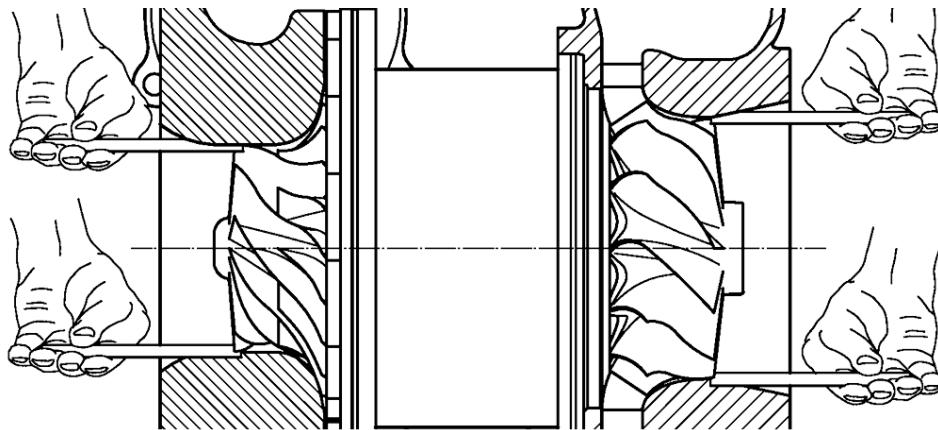
Components of T/C

The mean values measured must be within the permissible values of the radial clearance N and R.

Clearances out of tolerance and excessively worn components can be the cause of serious damage to machinery and property.

Measuring N and R

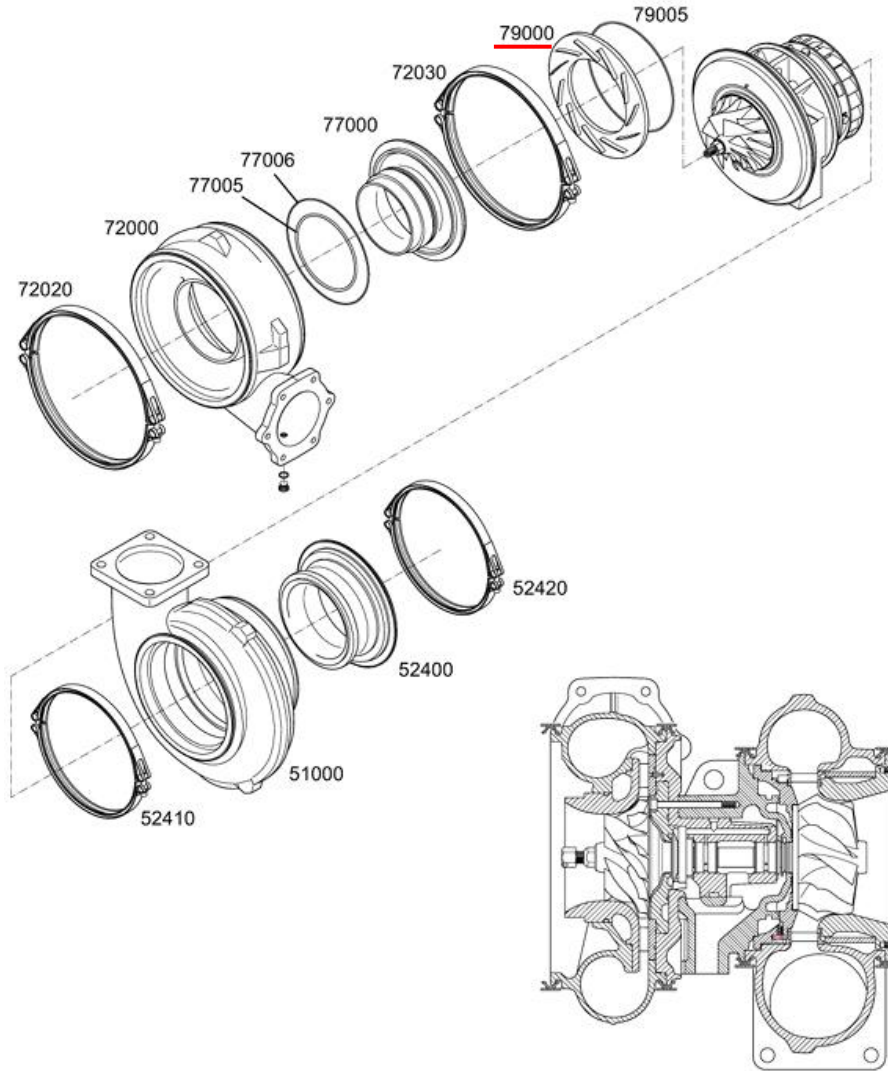
$$\frac{N1 + N2}{2} = N$$



$$\frac{R1 + R2}{2} = R$$

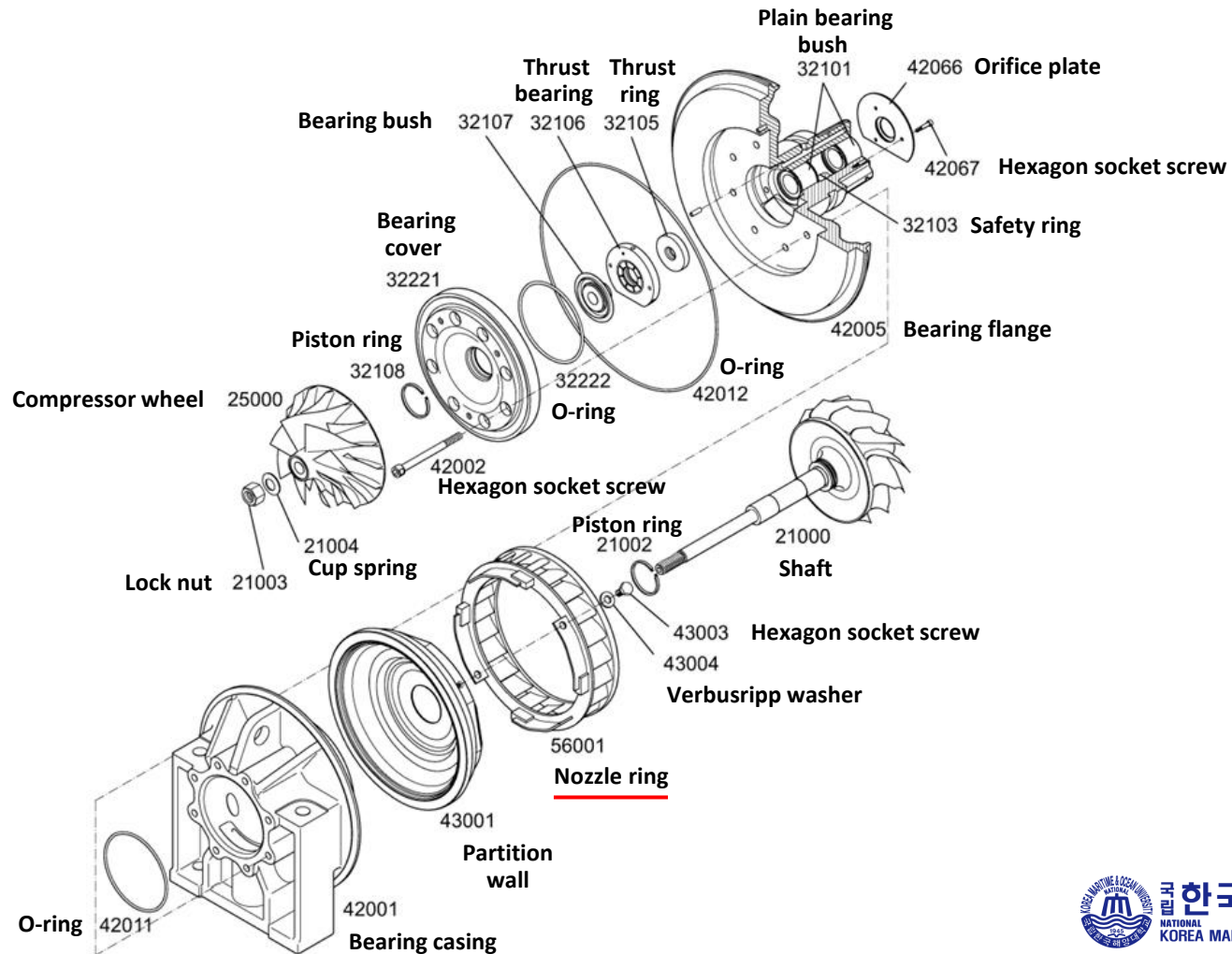
Radial clearances N and R
[mm]

	N	R
TPS 50	0.66...0.92	0.65...0.93



Part number	Description
51000	Turbine casing
52400	Gas outlet flange
52410	V-clamp
52420	V-clamp
72000	Compressor casing
72020	V-clamp
72030	V-clamp
77000	Wall insert
77005 (in spare parts kit)	O-ring
77006 (in spare parts kit)	O-ring
<u>79000</u>	<u>Diffuser</u>
79005 (in spare parts kit)	O-ring

12.3 View of cartridge group showing part numbers



Cleaning of T/C

Regular cleaning of the exhaust turbine is essential to ensure optimum engine operation using HFO as fuel.

Depending on the quality of the fuel used, particularly hard deposits can be formed on the nozzle ring in conjunction with high exhaust gas temperatures.

Heavily contaminated turbine components, in particular a contaminated nozzle ring, lead to deterioration in turbine efficiency. Reduced engine power and poorer efficiency are the result.

Under certain circumstances, even the omission of only one turbine washing cycle can lead to a build-up of carbon residues that can no longer be removed with the next regular cleaning process. The resulting need for an early turbocharger service increases the operating costs of the engine plant.



Contaminated
nozzle ring of T/C



Cleaning of T/C

Turbine cleaning

Generally, there are two different washing processes for cleaning the turbine: short-term and long-term washing. Which washing process to use depends on the turbocharger type.

In short-term washing, water is introduced into the turbine several times for a short time. This leads to temperature shocks which cause the dirt to flake off.

The cleaning effect of long-term washing is based on dissolving the dirt with water. Thus, the 10 minute cleaning time should ensure that all dirty surfaces are wetted with water and that dirt is dissolved.

Turbocharger Type	Washing process	Duration	Water quantity	Cleaning intervals
ABB TPS61-F	short-term washing	3x 30s	5 - 12 l / washing	150 – 200 h
ABB A145-M	long-term washing	10 min	12 l / min	50 – 200 h



Cleaning of T/C

Despite regular turbine washing in compliance with the washing parameters, if the turbines and nozzle rings are dirty, manual cleaning of the components is necessary.

Chemical analyses show that usually deposits on turbines and nozzle rings are mainly oxidic vanadium compounds. If fuels with a high vanadium content are used, we recommend shortening the washing intervals.

Field experience and tests have shown improved cleaning results with the long-term washing procedure, therefore in case operators experience insufficient cleaning results with the short term water injection, ABB recommends changing to long-term washing.

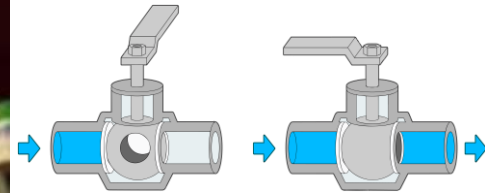
Dry cleaning of T/C turbine side



Water washing of T/C turbine side

<https://www.youtube.com/watch?v=vzn5oku3xOo>

T/C cleaning for compressor and turbine sides



Ball valve

T/C cleaning for compressor

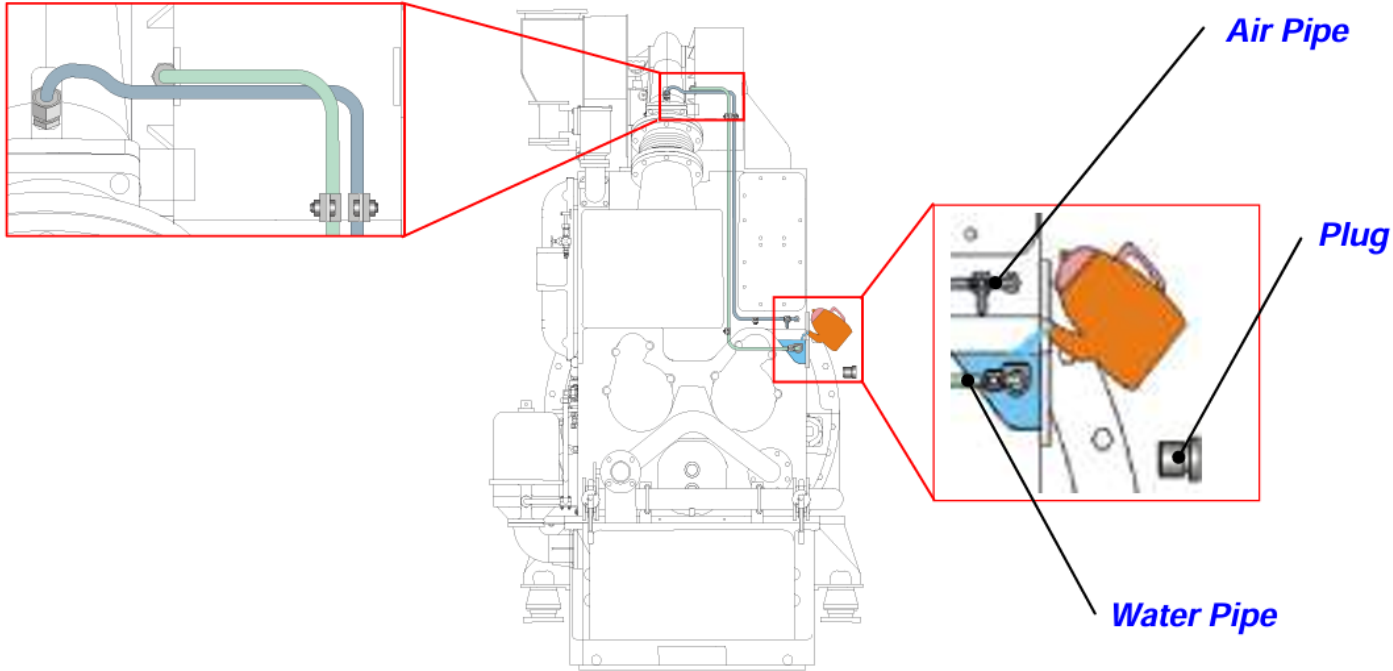


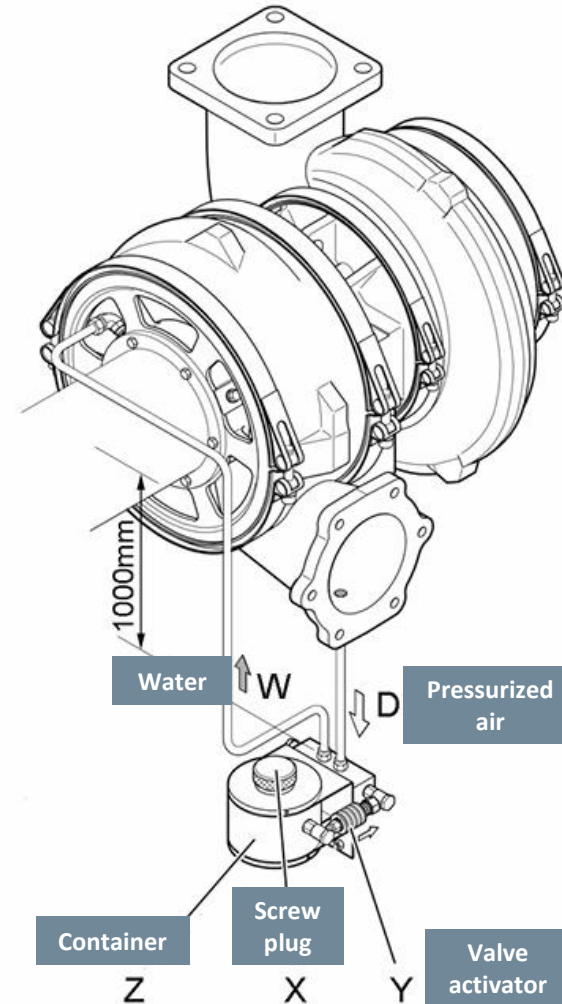
Fig.1 Pipe arrangement for cleaning the compressor side (압축기 세척용 파이프 배치상태)

T/C cleaning for compressor

Operating state prerequisites for cleaning compressor with XC1

To be able to carry out a satisfactory cleaning process which has been tested and is recommended by ABB Turbocharging, the following prerequisites must be fulfilled:

- 1) Run engine at load from 50 ... 85 %.
- 2) Start cleaning cycle according to following description Wet cleaning operation with XC1.



T/C cleaning for compressor

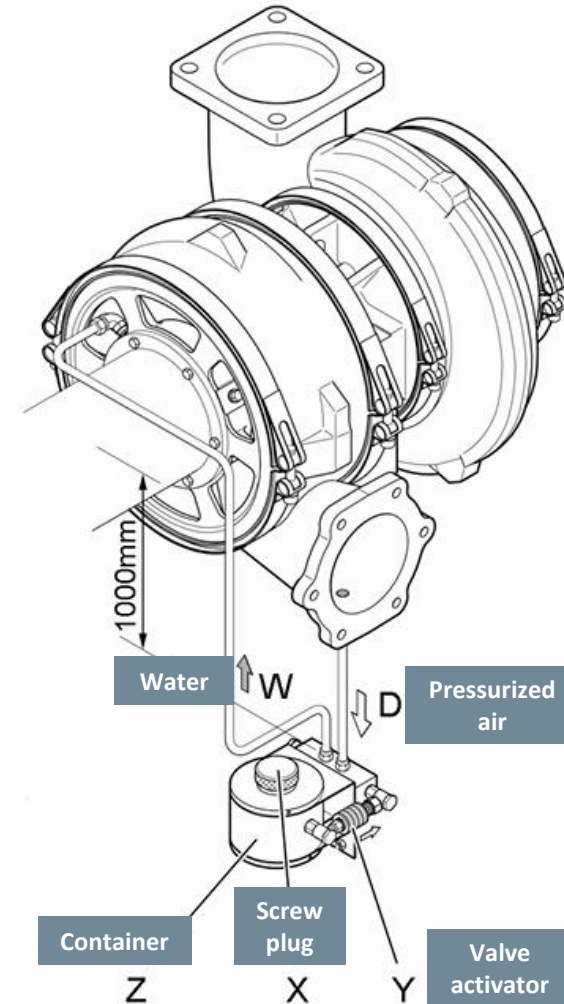
Operating state prerequisites for cleaning compressor with XC1

Operating the water pressure vessel

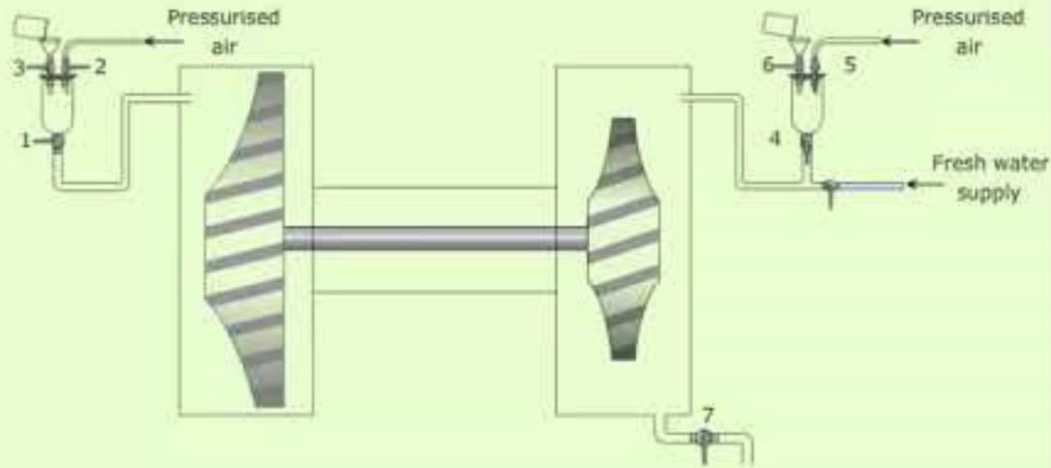
- 1) Remove the screw plug (X).
- 2) Fill the vessel with clean water.
- 3) Replace the screw plug.
- 4) Push the valve activator (Y) against the spring and hold it for 10 to 15 seconds until entire volume of water has been injected.
- 5) After cleaning, wait at least 5 minutes with the engine running to allow the turbocharger to dry.
- 6) The cleaning cycle may only be repeated after a stabilization period of at least 10 minutes has elapsed.

Cleaning parameters per turbocharger compressor

Turbocharger type	Turbocharger speed	Contents of water container [dm ³]	Water injection time t ₁ [s]
TPS 50	approximately n _{Bmax}	0.4	4 ... 10

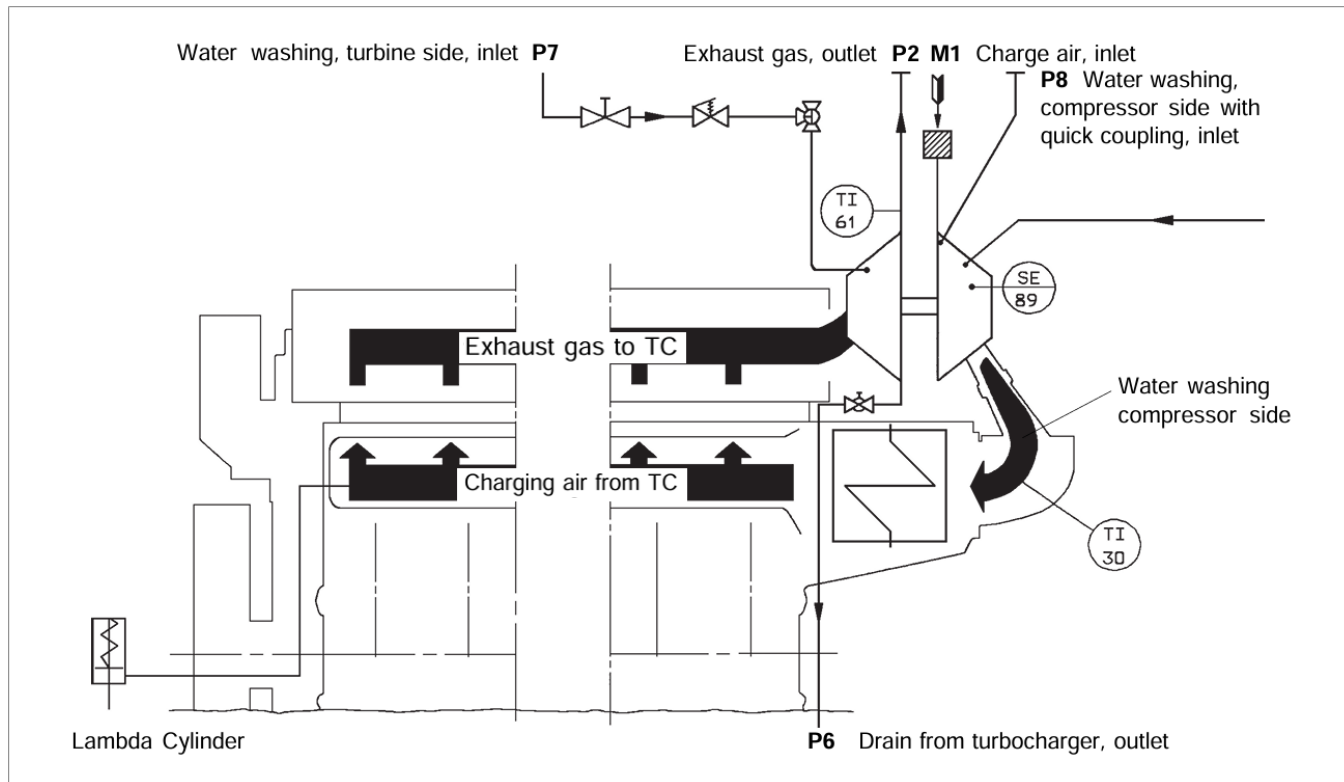


T/C cleaning for compressor and turbine sides



iv) Open the air cock 5 and wait for a minute.

T/C system



Troubleshooting for T/C

Malfunctions when starting

Sluggish start-up

Possible causes	Elimination
Turbocharger	Turbocharger contaminated
	Clean it
	Damaged bearing
	Contact an ABB Turbocharging Service Station
	Rubbing rotor
	Service Station
	Foreign object in turbocharger

Vibrations

Possible causes	Elimination
Turbocharger	Rotor unbalance
	Contact an ABB Turbocharging Service Station
	Damaged turbine or compressor
	Damaged bearing

Rubbing of rotating parts

Normal behaviour, no malfunction

- Turbocharger
- Minor uniform wear around the periphery of rotor components, caused by slight local rubbing of adjacent components, is permissible. The compressor blades and turbine blades are then shortened somewhat. Certain tolerances must be observed to avoid a significant loss of efficiency.
- If there is any doubt about the extent of rubbing, then an ABB Turbocharging Service Station must be contacted.
 - Have a dimension check carried out by an ABB Turbocharging Service Station.



Troubleshooting

Surging of the turbocharger

Turbocharger surging

Possible causes		Elimination
Engine	Guard in front of the turbocharger is contaminated or damaged	Clean / replace it
Turbocharger	Filter silencer or diffuser contaminated Heavy deposits of dirt in the turbine or nozzle ring	Clean it
Possible causes		Elimination
Charge air cooler	Cooler contaminated Charge air duct blocked	Clean it

⚠ CAUTION

Prolonged or periodic surging

Possible damage to components, such as the compressor wheel, turbine blades, bearings and filter silencer

- ▶ Have the cause clarified immediately by an ABB Turbocharging service station and rectified.
- ▶ Have components inspected for damage and, if necessary, replaced by an ABB Turbocharging service station.

Sporadic surge blows

Surging of the turbocharger can occur during certain operating conditions such as when reducing the engine performance quickly when manoeuvring. At the same time, the flow direction in the compressor is momentarily reversed. Such sporadic surge blows do not impair the safe operation of the turbocharger.

- A surge blow is accompanied by a loud bang and escape of hot air from the filter silencer.



T/C surging on G/E



Troubleshooting for T/C

Malfunctions during operation

Lubricating oil pressure too low

Possible causes		Elimination
Engine	Oil filter heavily contaminated	Clean it
	Defective oil pump in lubricating system	Check / replace it
	Manometer display wrong	Replace the manometer
Turbocharger	Rotor axial clearance too large	Contact an ABB Turbocharging Service Station

Reduction in speed

Possible causes		Elimination
Engine	Defects of connected cylinders when pulse charging	Contact the enginebuilder
Turbocharger	Heavy contamination of the turbine	Clean it
	Damaged rotor components or bearing	Contact an ABB Turbocharging Service Station
Pipes	Defects such as leaks in the exhaust gas pipes or charge air ducts	Make repairs



Troubleshooting for T/C

Increase in speed

Possible causes	Elimination
Turbocharger Heavily contaminated nozzle ring (with 4-stroke application)	Contact an ABB Turbocharging Service Station

Exhaust gas temperature too high

Engine performance and engine speed unchanged

Possible causes	Elimination
Engine Malfunctioning injection system	Repair it or contact the manufacturer
Turbocharger Air starvation, for example filter silencer clogged with dirt	Clean it
Compressor / turbine contaminated	
Exhaust counterpressure too high	Clean or repair the boiler or exhaust silencer
Enlarged flow cross-section of the nozzle ring due to deformed blade trailing edges	Contact an ABB Turbocharging Service Station
Damaged or eroded turbine	
Charge air cooler Cooler contaminated	Clean it
Cooling water volume too low	Top up the water
Entry temperature of cooling water too high	Inspect / clean the cooling system
Inadequate ventilation	Improve the ventilation



Troubleshooting for T/C

Charge air pressure too low

Engine performance and engine speed unchanged, air intake condition normal

Possible causes		Elimination
Engine	Air receiver leaking	Make repairs
	Gas piping between the engine and the turbine leaking	
	Poorly adjusted injection system	Correct it
	Poorly adjusted valve control	
Turbocharger	Manometer display wrong	Replace the manometer
	Line to manometer leaking	Repair the leak
	Contaminated filter silencer causing excessive loss of pressure	Clean it
	Compressor / turbine contaminated	
	Damaged compressor / turbine	Contact an ABB Turbocharging Service Station
	Exhaust counterpressure too high	Clean or repair the boiler or exhaust silencer

Charge air pressure too high

Engine performance and engine speed unchanged, air intake condition normal

Possible causes		Elimination
Engine	Malfunctioning injection system	Repair it or contact the manufacturer
	Poorly adjusted injection system	Correct it
	Engine performance higher than expected	Check engine performance
Turbocharger	Manometer display wrong	Replace the manometer



Troubleshooting for T/C

Malfunctions when stopping

Noises during run-down

Possible causes	Elimination
Turbocharger	Turbocharger contaminated
	Clean it
	Damaged bearing
	Contact an ABB Turbocharging Service Station
Rubbing rotor	
	Foreign object in turbocharger

Run-down time too short

Possible causes	Elimination
Turbocharger	Turbocharger contaminated
	Clean it
	Damaged bearing
	Contact an ABB Turbocharging Service Station
Rubbing rotor	
	Foreign object in turbocharger



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